

ADAM GARDNER

LEAD DATA SCIENTIST, ARTINUS CONSULTING INC.

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DEEP LEARNING, LLMS & AI AGENTS | PHYSICS-INFORMED NEURAL NETWORKS | AI SYSTEMS & DEPLOYMENT

HIGHLIGHTS OF QUALIFICATIONS

- Lead Data Scientist and co-founder at Artinus Consulting Inc.
- Adjunct Professor of Computing and Data Analytics at Saint Mary's University
- Ph.D. in Mathematical Physics from University of Toronto
- 5+ years experience in designing and deploying ML and AI systems
- Expert in AI agents, LLM frameworks, distributed computing and full-stack data science pipelines

PROFESSIONAL EXPERIENCE

REVELA SYSTEMS | VICTORIA, BC | FEBRUARY 2025 – JULY 2025

Opus (February 2025 – July 2025)

- Fine-tuned open-source LLMs with Unsloth and Axolotl for agentic workflows.
- Used Outlines and XGrammar to guide LLMs to output a custom JSON schema.
- Used multi-GPU distributed computing to train and serve extremely large models using AWS EC2.
- Deployed production-ready LLMs on AWS using Docker, Kubernetes, FastAPI and vLLM.

CANADA BORDER SERVICES AGENCY | OTTAWA, ON | DECEMBER 2021 – MARCH 2025

Security Screening Automation (December 2021 – March 2025)

- Collaborated with a diverse team of data scientists, IT professionals, cloud developers and business managers to modernize border screening software.
- Used Amazon SageMaker on Amazon Web Services (AWS) to train and deploy machine learning models and LLMs.
- Performed entity resolution using Named Entity Recognition (NER), fuzzy search and LLMs (Llama)
- Improved upon the Levenshtein edit distance for multi-word matches using a novel log-weighting method and a light-weight ML classifier.
- Used LLMs (Mistral) to create synthetic documents to augment real data.
- Applied Optical Character Recognition (OCR) models for structured data extraction from scanned documents.

ARTINUS CONSULTING | ST. JOHN'S, NL | APRIL 2021 – PRESENT

SKU Pilot (May 2023 – Present)

- Managed a team of data scientists to develop a logistics optimization app.
- Deployed AI Agents on Azure for user documentation RAG, SQL generation and API orchestration.
- Created complex dynamic SQL queries and API endpoints.
- Collaborated on a proprietary time series forecasting model and AutoML pipeline.
- Applied mixed-integer linear programming to minimize business costs while satisfying business constraints.
- Implemented just-in-time (JIT) purchase order scheduling using a greedy algorithm.
- Discovered new scaling laws for AutoML pipelines and time series forecasting metrics.

Physics-Informed Neural Networks for Electromagnetic Simulation and Inverse Design (March 2024 – Present)

- Co-supervised R&D of Physics-Informed Neural Networks (PINNs) for numerically solving Maxwell's equations.
- Developed novel neural network architectures which encode symmetries of the parametric wave equation.

Data Efficient Medical Imaging Vision Transformers (August 2023 – January 2025)

- Developed novel vision transformer architectural modifications on medical imaging data.

- Mentored masters students on best practices such as modular programming, unit testing and version control.

Vision Transformer Architecture Research (July 2022 – January 2025)

- Invented a new vision transformer architecture with translation equivariance and linear scaling with context length.
- Created a new transformer positional encoding technique which increased CFAR-100 top-1 accuracy by 15% compared to traditional positional encoding methods (79% vs 64%).

Transport Canada (September 2021 – Present)

- Translated business requirements from economics subject matter experts into Python-based analytical models.
- Researched and developed a novel methodology for synthesizing inconsistent data from different sources.
- Wrote Python and SQL code and used various APIs to consolidate several large databases.
- Improved upon the iterative proportional fitting (IPF) algorithm to infer unknown values from aggregate totals.

TORUSOFT | HALIFAX, NS | FEBRUARY 2021 – APRIL 2023

Defence Research and Development Canada (April 2021 – April 2023)

- Deployed state-of-the-art transformer models for language detection and translation between 9,900 language pairs.
- Developed a distributed ETF and modeling pipeline leveraging Python, Docker, Spark and MLFlow.
- Applied Belief Propagation and network physics algorithms to model the spread of information.
- Created and deployed Business Intelligence (BI) dashboard views for analysts.
- Classified documents using the unsupervised Top2Vec algorithm.

UNIVERSITY OF TORONTO | TORONTO, ON | SEPTEMBER 2015 - AUGUST 2019

Teaching Assistant

- Instructed 15 courses in mathematics and statistics for 30 months, assisting hundreds of students per course.
- Consistently rated 4 to 5 out of 5 by course feedback forms for clear, insightful explanations of course materials.

EDUCATION AND DESIGNATIONS

Saint Mary's University, Halifax, NS: 2024

Adjunct Professor of Computing and Data Analytics

- **Relevant Projects:** SKU Pilot, Physics-Informed Neural Networks for Electromagnetic Simulation and Inverse Design, Data Efficient Medical Imaging Vision Transformers, Vision Transformer Architecture Research

University of Toronto, Toronto, ON: 2020

Ph.D. Mathematics (GPA: 3.88)

- **Relevant Coursework:** Massive Data Analysis, Applied Nonlinear Equations, Partial Differential Equations, Galois Theory, Real Analysis, Quantum Field Theory
- **Awards/Honors:** Malcolm Slingsby Robertson Fellowship in Mathematics
- **Relevant Projects:** Thesis: *Instability of Electro-Weak Homogenous Vacua in Strong Magnetic Fields*

University of Toronto, Toronto, ON: 2015

Master of Science – Mathematics (GPA: 3.80)

- **Relevant Coursework:** Topology, Perturbation Theory, Set Theory, Complex Analysis, Algebra
- **Awards/Honors:** NSERC CGSM Scholarship
- **Relevant Projects:** Thesis: *The Stability Theorem of Persistent Homology*

Memorial University of Newfoundland, St. John's, NL: 2014

Bachelor of Science – Applied Math and Physics (Honours; GPA: 3.97)

- **Relevant Coursework:** Partial Differential Equations, Ring Theory, Algebraic Topology, Functional Analysis, Geometric Group Theory, Differential Geometry, Number Theory, Quantum Mechanics, General Relativity

- **Awards/Honors:** SHAD Entrance Scholarship, Williams Science Scholarship
- **Relevant Projects:** Thesis: *A Combinatorial Riemannian Penrose Inequality for Discrete Anti-de Sitter Space*

PUBLICATIONS

- Gardner - *The stability theorem of persistent homology* (Morfismos, Vol. 21, No. 1, 2017, pp. 15–43)
- Gardner & Sigal - *Instability of electroweak homogeneous vacua in strong magnetic fields* (Ann. Henri Poincaré 2024)
- Andrew Harris, Vineeth Aljapur, Jason Carayanniotis & Adam Gardner - *FAST: Filamentous Actin Segmentation Tool for quantifying cytoskeletal organization* (Journal of Cell Science 2025)
- Adam Gardner & Aria Sardari - *FViT: Fourier Transform-based Positional Encodings for Vision Transformers* (preprint)

ADDITIONAL CREDENTIALS

TECHNICAL SKILLS	▪ LLM Models & Services: Mixtral 8x7B, LLaMA 7B, DeepSeek R1, ChatGPT 4o ▪ Agent Frameworks & Orchestration: LangChain, MCP ▪ Retrieval-Augmented Generation (RAG): FAISS, Pinecone, LangChain ▪ LLM Serving: vLLM, Ollama ▪ LLM Fine-Tuning: PEFT, Axolotl, Unsloth, LoRA ▪ Structured Generation & Control: Outlines, XGrammar ▪ LLM Performance Optimization: Quantization, Speculative Decoding, KV Caching ▪ Experiment Tracking & Reproducibility: MLflow, DVC, Docker ▪ Neural Network Architectures: CNNs, GNNs, RNNs, LSTMs, Transformers ▪ Deep Learning: PyTorch, TensorFlow, Hugging Face ▪ Classical ML: SK-Learn, CatBoost, Time-Series Forecasting, AutoML ▪ Topic Modeling: Top2Vec, LDA, LLM Prompt Engineering ▪ Graph algorithms: PageRank, Belief Propagation, NetworkX, Neo4j, GNNs ▪ Scientific ML: Physics-Informed Neural Networks (PINNs), Operator Learning ▪ API & Service Development: FastAPI, RESTful services ▪ Distributed Computing: Spark, Kubernetes, AWS EC2 ▪ Databases & Storage: PostgreSQL, Cassandra, Pinecone, Neo4j ▪ Programming Languages: Python, SQL, C++, Java, JavaScript, CSS, Fortran ▪ Data Analysis & Scientific Tools: Pandas, Polars, NumPy, R, MATLAB, Mathematica ▪ Technical Communication & Dashboards: Jupyter, Marimo, LaTeX, Plotly Dash ▪ Productivity Tools: Microsoft Word, Excel, PowerPoint, LLMs
HONORS & AWARDS	▪ 1st Place, W.J. Blundon Contest – Memorial University of Newfoundland (2009) ▪ 3rd Place, Science Atlantic Math Competition – Science Atlantic (2011 & 2013) ▪ Black Belt, Kenpo Karate – Rock Athletics (2013)
PROFESSIONAL DEVELOPMENT	▪ Machine Learning – Stanford University (Coursera 2015) ▪ Advanced Machine Learning Specialization – Moscow University (Coursera 2021) ▪ PySpark Essentials for Data Scientists (Big Data + Python) – Udemy
ORGANIZATIONS	▪ Toronto Machine Learning Summit ▪ Toronto Synthetic Intelligence Forum
INTERESTS	Video Game Development & Speedrunning, Martial Arts (Kenpo Karate, Jiu-Jitsu, Kickboxing), Competitive Bodybuilding & Powerlifting, Guitar & Performing Music